

cuOpt benchmark: GS Retail routing

T2 · Appendix A1–A2 · CVRPTW · 537–586 orders

Result: cuOpt returns **2,610 km at 60 seconds**, compared with 2,815 km for OR-Tools: approximately **7.3% shorter aggregate distance** on the reported three-instance comparison.

Omelet technical report · measured engine performance. This report documents the experiments presented in the final-pitch appendix: measured results, solver comparisons, test conditions and interpretation. Public results are reported by Omelet; customer-level inputs and internal run records are not reproduced here. Benchmark execution is distinct from the NVIDIA integration roadmap.

Metric	Route distance (km) and per-instance gap to the best result among compared solvers; 10- and 60-second budgets.
Problem	Three GS Retail instances; A3 replenishment scenario; 537–586 orders.
Setup	Full-results appendix: one RTX A6000, August 2026, drive-distance objective and 55 km/h. Summary appendix also mentions RTX 4090 / A6000. Repeated runs and re-validation are reported; per-run device mapping/repetition count are not itemized in the public appendix.
Date / reporting entity	Experiment: August 2026. Report issuer: Omelet. The public appendix does not name the individual experiment owner.

METHOD	10 S: KM	10 S: GAP	60 S: KM	60 S: GAP
NVIDIA cuOpt	2,684	+1.09%	2,610	+0.83%
Omelet RouteOpt	2,676	+1.37%	2,622	+1.70%
RouteOpt, ALNS configuration	2,984	+12.22%	2,766	+7.16%
OR-Tools	3,000	+12.79%	2,815	+9.50%
LKH-3	No feasible solution	at 1,830 s	≈30× the 60 s budget	—

How to interpret the results

Gap uses a moving, per-instance best-across-solvers reference. Compare distance across time budgets, not the gap percentages as if they shared a fixed global optimum. Aggregated km and mean per-instance gap may rank differently; raw per-instance records are needed to reconstruct the averages.

The appendix reports 2/3 cuOpt wins at both budgets and run-to-run variation: cuOpt ±1.24 / ±0.50 percentage points and RouteOpt-ALNS ±0.87 / ±1.27 percentage points (10 / 60 seconds). The deck describes the frontier difference as within noise. A formal significance/equivalence test and comparable variation for primary RouteOpt are not supplied; no statistical-tie conclusion is independently verified here.

Technical interpretation

Distance reduction versus OR-Tools is calculated as $(2,815 - 2,610) / 2,815 = 7.28\%$. This is a solver comparison on the tested instances, not a customer cost-savings measurement. Customer orders and routes remain confidential.

[Open appendix A2: full results](#) · [Read the consolidated technical report](#)